

DATA ENGINEERING TRAINING REPORT (WEEK 1 – WEEK 6)

1. Introduction

During the six weeks of Data Engineering training, I was introduced to the core concepts and practical tools required in modern data systems. The training focused on building a strong foundation in data pipelines, data processing, data modeling, storage systems, and transformation techniques.

Through structured weekly tasks, I gained both theoretical understanding and hands-on experience with industry-standard tools such as Python, SQL, PostgreSQL, dbt, and cloud data warehouse concepts. This training helped me understand how raw data is converted into meaningful insights through well-designed data engineering workflows.

2. Weekly Work Summary

2.1 WEEK 1: Data Engineering Foundations and Environment Setup

During the first week of the training program, I focused on understanding the fundamental concepts of Data Engineering and preparing a complete development environment required for upcoming practical tasks.

I studied how modern data systems are structured, starting from data generation at source systems to ingestion, storage, transformation, and final data consumption in analytics and dashboards. This helped me understand the complete lifecycle of data and how it flows through different stages in a real-world data pipeline.

I also learned the differences between various data-related roles such as Data Engineer, Data Analyst, Analytics Engineer, and Machine Learning Engineer. This helped me clearly understand their responsibilities and how each role contributes to the overall data ecosystem.

Additionally, I explored key architectural concepts such as batch processing and streaming systems, which explained how data is processed either in large batches or in real-time. I also studied foundational data system concepts such as OLTP (Online Transaction Processing),

OLAP (Online Analytical Processing), data warehouses, data lakes, and lakehouse systems, which are widely used in modern data architectures.

For practical implementation, I set up a complete development environment required for data engineering tasks. This included configuring necessary software tools, setting up project folders, initializing version control, and preparing a structured workspace for all upcoming assignments. I also created a basic data architecture diagram to visually understand how data moves from source systems through pipelines into storage and analytics layers.

This setup phase helped me build a strong foundation for performing future practical tasks efficiently and following a structured development workflow.

2.2 WEEK 2: SQL for Data Engineering

During the second week, I focused on developing strong SQL skills required for data transformation and analytics.

I studied relational database concepts and practiced advanced SQL techniques including joins, subqueries, CTEs, aggregation, conditional logic, and set operations. A major focus was on window functions such as ranking, running totals, and deduplication.

I also studied sessionization, Slowly Changing Dimensions (SCD), and query optimization techniques using execution plans.

For practical work, I executed complex SQL queries on sample datasets, performed analytical operations, and analyzed query performance using database tools.

Tools Used:

- PostgreSQL
 - pgAdmin
 - SQL (Query Language)
 - EXPLAIN ANALYZE (PostgreSQL tool)
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2.3 WEEK 3: Python for Production Data Pipelines

During the third week, I focused on building production-level Python skills for data engineering workflows.

I studied Python project structure, object-oriented programming, modular programming, and data pipeline design principles. I also learned key libraries used in data engineering such as Requests,

Pandas, SQLAlchemy, and Pydantic. In addition, I studied logging, error handling, retry mechanisms, and testing frameworks.

For practical work, I built a complete API-based data pipeline that extracted, validated, transformed, and stored data. I also implemented logging, retries, and automated tests to improve reliability.

Tools Used:

- Python 3.11
 - VS Code
 - Requests library
 - Pandas
 - Pydantic
 - SQLAlchemy
 - Pytest
 - JSON / CSV file system
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2.4 WEEK 4: Data Modeling for Analytics

During the fourth week, I focused on data modeling techniques used in analytical systems.

I studied normalization concepts (1NF, 2NF, 3NF) and dimensional modeling, including fact tables, dimension tables, and schema design (star and snowflake schemas). I also studied Slowly Changing Dimensions (SCD Type 2) and how historical data is maintained in data warehouses.

For practical work, I designed a complete data model for a selected business domain, created an ERD diagram, and defined SQL table structures with proper relationships and grain definition.

Tools Used:

- PostgreSQL
 - SQL (DDL)
 - Draw.io / ERD diagram tools
 - Database schema design tools
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2.5 WEEK 5: Cloud Warehouse and Object Storage

During the fifth week, I studied cloud-based data storage systems and modern data warehouse architectures.

I learned about platforms such as Snowflake and BigQuery and their separation of compute and storage. I also studied object storage systems, partitioning, clustering, file formats (CSV, JSON, Avro, Parquet), and security concepts like IAM and access control.

For practical work, I converted raw data into Parquet format, organized it into partitioned folder structures, and loaded it into warehouse-style tables. I also performed analytical queries to understand performance differences.

Tools Used:

- Python
 - Pandas
 - PyArrow
 - Parquet file format
 - PostgreSQL / Warehouse simulation
 - Snowflake (theoretical)
 - BigQuery (theoretical)
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2.6 WEEK 6: Data Transformation with dbt

During the sixth week, I focused on implementing data transformation pipelines using dbt (Data Build Tool).

I studied dbt architecture, including staging, intermediate, and mart layers. I also learned about dbt features such as refs, sources, lineage, macros, materializations, and testing frameworks.

For practical work, I built a complete dbt project connected to a database. I created staging models, intermediate transformations, and final mart models. I also implemented data tests such as not-null, unique, relationships, and accepted values. Finally, I generated dbt documentation and analyzed data lineage.

Tools Used:

- dbt Core
 - PostgreSQL
 - SQL
 - Command Line Interface (CLI)
 - dbt documentation tool
 - Git / GitHub
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3. Overall Summary

Throughout the six weeks of training, I developed a strong foundation in Data Engineering, covering data ingestion, transformation, modeling, storage, and analytics.

I gained hands-on experience with SQL, Python, dbt, and database systems while also learning modern data architecture concepts and cloud data platforms. This training helped me understand how real-world data pipelines are designed and implemented in production environments.

The training significantly improved my technical knowledge, problem-solving ability, and understanding of end-to-end data workflows.

4. Conclusion

The six-week Data Engineering training program was a highly valuable learning experience. It provided both theoretical knowledge and practical exposure to essential data engineering tools and concepts.

I now have a clear understanding of how data flows from source systems to analytics platforms through structured pipelines. This experience has strengthened my confidence and prepared me for real-world data engineering projects.